Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

Date:30-7-26

AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.
Objective:
1. To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
2. To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.
3. To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

Procedure

1. Define soil moisture range (0–100%) and set an irrigation threshold value (e.g., 40%).
2. Generate or input soil moisture data to simulate field conditions.
3. Compare soil moisture with the threshold value to determine pump status (ON if low moisture, OFF if sufficient moisture).
4. Simulate water pumping action based on control logic.
5. Calculate irrigation efficiency based on moisture improvement and water usage.
6. Display soil moisture level, pump status, and efficiency results.
7. Plot graphs to analyze soil moisture variation and irrigation performance.
8. Evaluate system behavior for smart agriculture decision-making.

Objective 1

Matlab Code and Output:

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% -------- Graph 1: Soil Moisture Variation --------
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Moisture Status (Dry/Wet Level) --------
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on
```

EXP111.m

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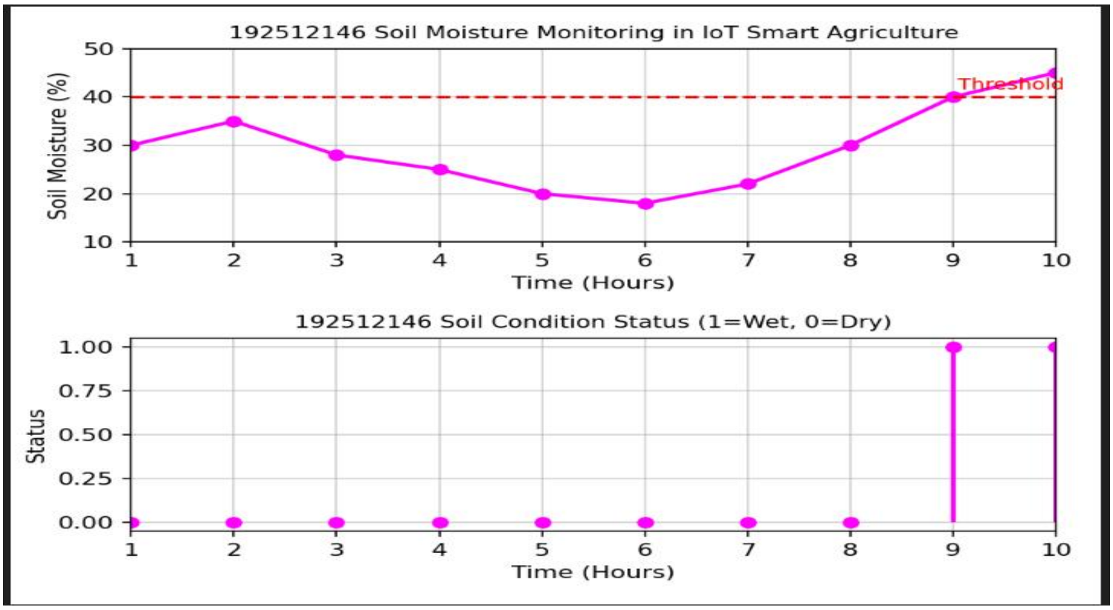
```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% -------- Graph 1: Soil Moisture Variation --------
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2,'Color',"magenta")
title('192512146 Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')

grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Moisture Status (Dry/Wet Level) --------
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2,'Color',"magenta")
title('192512146 Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
```



Soil moisture levels vary over time, showing fluctuations in field water content.
The threshold line helps identify dry and wet conditions for smart irrigation decisions.

Objective 2

Matlab Code and Output:

```
clc; clear; close all;
% Simulated soil moisture (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
```

% -------- Graph 1: Soil Moisture vs Pump Status --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Pump Status (ON/OFF) --------
subplot(2,1,2)
stem(time, pump_status, 'filled','LineWidth',2)
title('Water Pump Status (1=ON, 0=OFF)')
xlabel('Time (Hours)')
ylabel('Pump Status')
grid on

exp112.m

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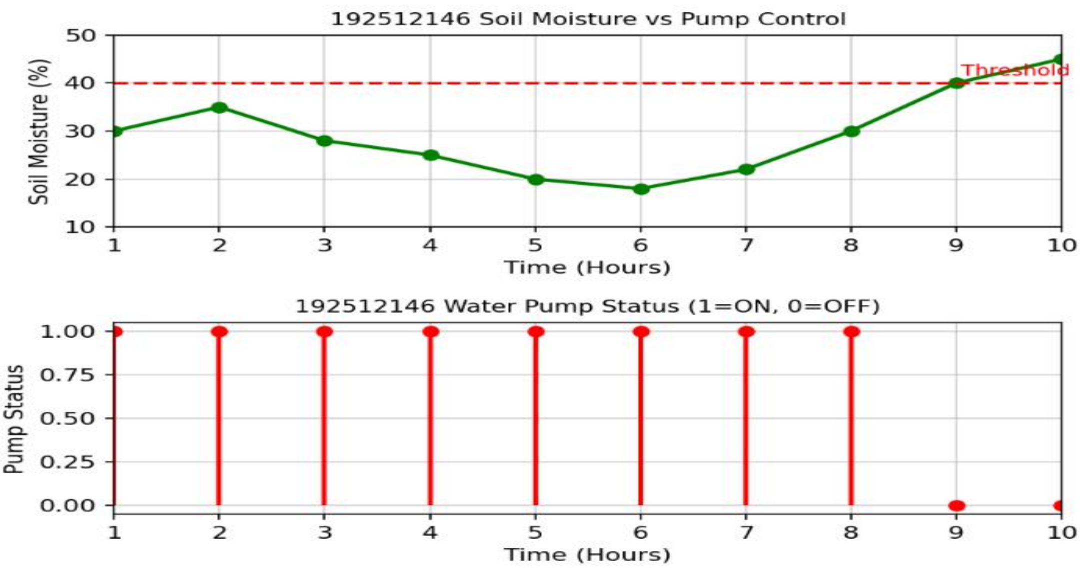
```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
% -------- Graph 1: Soil Moisture vs Pump Status --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2,'Color',"g")
title('192512146 Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Pump Status (ON/OFF) --------
subplot(2,1,2)
stem(time, pump_status, 'filled','LineWidth',2,'Color',"r")
title('192512146 Water Pump Status (1=ON, 0=OFF)')
xlabel('Time (Hours)')
```



The pump turns ON automatically when soil moisture drops below the threshold level.

When moisture is sufficient, the pump remains OFF, ensuring efficient water usage and smart irrigation control.

Objective 3

Matlab Code and Output:

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;
efficiency(efficiency > 100) = 100; % limit to 100%
figure
% -------- Graph 1: Soil Moisture vs Efficiency --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Irrigation Efficiency --------
subplot(2,1,2)
plot(time, efficiency, '-s','LineWidth',2)
title('Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
grid on
```

```
exp113.m
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clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;
efficiency(efficiency > 100) = 100; % limit to 100%
figure
% -------- Graph 1: Soil Moisture vs Efficiency --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2,'Color',"g")
title('192512146 Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Irrigation Efficiency --------
subplot(2,1,2)
plot(time, efficiency, '-s','LineWidth',2,'Color',"r")
title('192512146 Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
grid on
```

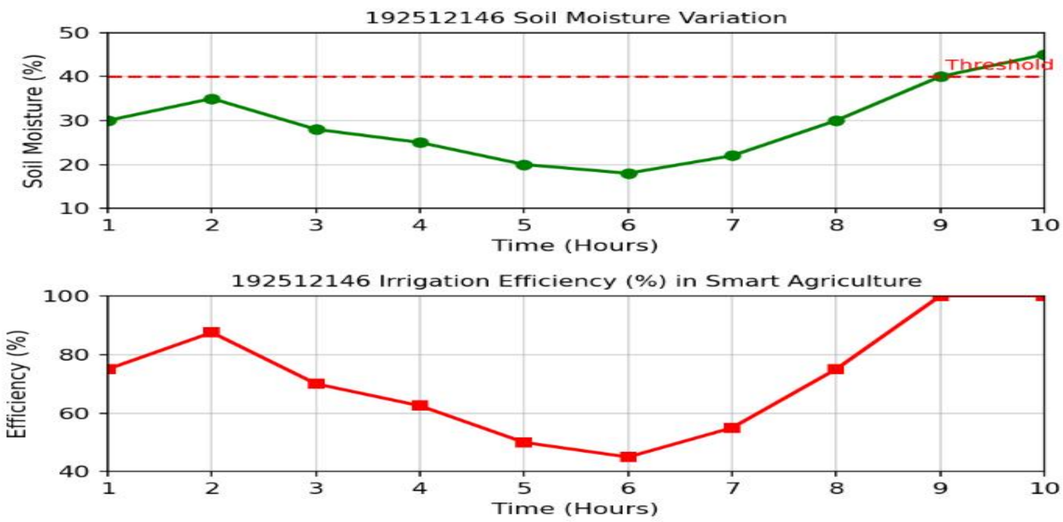

Irrigation efficiency increases when soil moisture approaches the optimal threshold level. The system ensures better water utilization by maintaining soil moisture within the desired range
using smart control.

Result:

1. The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.
2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.
3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level.